Software Requirements Specification

Inventory & Supply Chain Management System

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1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the requirements for the Inventory and Supply Chain Management System. The system automates order handling, logistics assignment, delay detection, and reassignment processes.

1.2 Scope

The system manages the full lifecycle of supply chain and inventory processing including:

- Order creation
- Inventory verification
- Warehouse allocation
- Carrier assignment
- Delay detection
- Automatic reassignment
- Dashboard monitoring

1.3 Definitions and Acronyms

Term	Meaning
SRS	Software Requirements Specification
API	Application Programming Interface
DB	Database
ADS	Automated Decision System
SLA	Service Level Agreement

1.4 Overview

The document defines system functionality, constraints, and operational requirements.

2 Overall Description

2.1 Product Perspective

The system is a supply chain automation platform that receives orders via API and processes them through automated workflows.

2.2 Product Functions

- Order creation and storage
- Inventory validation
- Warehouse selection
- Carrier assignment
- Delay detection engine
- Automatic reassignment
- Order lifecycle tracking
- Dashboard monitoring
- Logging and auditing

2.3 User Classes

2.3.1 Admin

- Creates and monitors orders
- Views dashboard
- Can override system decisions

2.3.2 Customer

- Places orders
- Tracks order status

2.3.3 Automated System

- Performs delay detection
- Executes reassignment
- Maintains workflow states

2.4 Operating Environment

- Web-based dashboard
- REST API backend
- SQL/NoSQL Database
- Automated workflow engine

2.5 Constraints

- Must support concurrent order processing
- Must maintain data consistency
- Must log workflow decisions

3 System Workflow

1. Order received via API
2. Order stored in database
3. Inventory and logistics assigned
4. Delay flags evaluated
5. Reassignment triggered if delay detected
6. Dashboard updated

4 Functional Requirements

4.1 Order Creation

The system shall allow Admin or Customer to create orders with:

- Product details
- Quantity
- Delivery location
- Expected delivery time

4.2 Inventory Check

The system shall:

- Verify product availability
- Reject order if stock is insufficient
- Proceed if stock is available

4.3 Warehouse Allocation

The system shall assign the nearest or most suitable warehouse.

4.4 Carrier Assignment

The system shall assign a delivery carrier automatically.

4.5 Delay Detection

The system shall periodically check delivery status using time thresholds and carrier flags.

4.6 Automatic Reassignment

If delay is detected:

- Reassign warehouse or carrier
- Update order state to REASSIGNED

4.7 Order State Management

The system shall maintain states:

- CREATED
- ASSIGNED
- IN TRANSIT
- DELAYED
- REASSIGNED
- DELIVERED

4.8 Logging and Traceability

The system shall log all workflow decisions and timestamps.

4.9 Dashboard Monitoring

Admin shall be able to monitor order lifecycle and reassignment history.

5 Non Functional Requirements

5.1 Performance

- Must process multiple orders concurrently
- Background jobs must execute periodically

5.2 Reliability

The system must ensure consistent workflow states.

5.3 Scalability

The architecture must support increasing order volumes.

5.4 Security

- Role-based authentication
- Secure APIs
- Data encryption

5.5 Availability

System should maintain high uptime.

5.6 Maintainability

System must support modular debugging and updates.

6 External Interface Requirements

6.1 User Interface

Dashboard shall provide:

- Order status visualization
- Delay alerts
- Reassignment history
- Order filtering

6.2 API Interface

System shall expose APIs for:

- Order creation
- Status retrieval
- Logistics updates

6.3 Database Interface

Database shall store:

- Orders
- Inventory
- Warehouse data
- Carrier details
- Logs

7 Use Case Model

7.1 Create Order

Actor: Admin
Outcome: Order saved with CREATED state

7.2 Process Order

Actor: System
Outcome: Order moves to ASSIGNED

7.3 Detect Delay

Actor: System
Outcome: Delay identified or cleared

7.4 Reassign Logistics

Actor: System
Outcome: Order reassigned automatically

7.5 Monitor Order

Actor: Admin
Outcome: Full order lifecycle visibility

8 System Constraints

- Requires carrier status feeds
- Requires real-time inventory updates
- Requires stable network connection

9 Future Enhancements

- AI-based predictive delay detection
- GPS tracking integration
- Customer notification service
- Dynamic carrier optimization

10 Appendix

The system supports automated delay handling, concurrency, inventory validation, reassignment workflows, and scalable supply chain automation.